

Legal Ease

Dynamic Previews and Export Options

Feature specification for live document previews and multi-format export

Document type	Feature specification
Version	1.0 (Draft)
Status	For review

1. Overview

Lawyers need to see exactly what a document will look like before it goes to a client, a court or the other side. Dynamic previews show a live, accurate rendering that updates as data, templates or settings change. Export options then deliver the same output in the format the recipient needs, with the right controls for privilege and security.

1.1 Goals

- Show what you will send before you send it, with no surprises in layout or merged data.
- Cut rework by catching missing fields and errors during preview.
- Offer the export formats law firms and courts actually require.
- Protect confidential and privileged content at the point of export.

1.2 Scope

Item	Previewable	Exportable
Engagement letters and fee agreements	Yes	PDF, DOCX, e-sign
Drafts from templates (pleadings, contracts, letters)	Yes	PDF, DOCX
Invoices and statements	Yes	PDF, CSV, XLSX
Court filings and exhibits	Yes	PDF, PDF/A, ZIP bundle
Reports and dashboards	Yes	PDF, XLSX, CSV
Matter files and correspondence	Yes	ZIP, PDF portfolio
Time entries and ledgers	Table view	CSV, XLSX

2. Dynamic Previews

2.1 Preview Types

Preview	Description
Live template preview	Choose a template and matter; the document renders with real data and updates as fields change.
Merge field highlighting	Merged values are highlighted; empty or invalid fields are flagged with a warning list.
Side-by-side edit	Form or data panel on one side, rendered document on the other, updating as you type.
Version comparison	Two versions shown together with additions and deletions marked.

Preview	Description
Client view	Shows the document as the client will see it in the portal, including redactions and permissions.
Print and page preview	Page breaks, headers, footers, margins and numbering as they will appear on paper or in PDF.
Bulk preview	Step through documents generated for many matters or clients before a batch run.

2.2 Functional Requirements

- Preview refreshes within 1 second of a field change for standard documents and shows a loading state for larger ones.
- Zoom, page thumbnails, page navigation and full-screen mode.
- Validation panel lists missing required fields, broken references and formatting issues, each linking to the source.
- Choose a data scenario (for example, individual vs company client) to test conditional sections in templates.
- Preview never changes stored data; edits are applied only when the user saves.
- Preview of exhibits and attachments with page count and total size.
- Read-only preview links for internal reviewers, with comments.

3. Export Options

3.1 Formats

Format	Use case
PDF	Standard delivery to clients, courts and counterparties
PDF/A	Long-term archival and court systems that require it
DOCX	Editable copies for negotiation and redlining
XLSX / CSV	Time, billing and report data for analysis and accounting
ZIP bundle	Full matter file or selected documents with folder structure and index
Print	Direct print with selected page range, duplex and tray options
Email and portal	Send the export directly, or publish to the client portal
E-signature	Send the exported PDF for signature and store the signed copy

3.2 Export Settings

- Page size, orientation, margins, header and footer text, and page numbering.
- Watermarks such as Draft, Confidential or Privileged, applied to selected pages.

- Bates numbering and exhibit stamping for litigation productions.
- Redaction with permanent removal of the underlying text, plus a redaction log.
- Metadata scrubbing for author, comments, tracked changes and hidden text.
- Table of contents, bookmarks and combined documents in a chosen order.
- Password protection and permission flags (no copy, no edit) for PDFs.
- Filename patterns such as MatterNo_ClientName_DocType_Date.

3.3 Batch and Scheduled Exports

- Export many documents or matters in one action, processed in the background with a progress view and email notification.
- Scheduled exports for recurring reports and invoices, delivered to email or cloud storage.
- Saved export presets per practice area, user or template.
- Expiring, permission-checked download links for large files.

4. User Flow

Step	User action	System behavior
1	Select document, template or report	Load data and render the first preview
2	Adjust data, options or template	Update the preview live and refresh the validation list
3	Resolve warnings	Link to each issue; clear warnings as fixed
4	Choose Export	Show format, settings, watermark and redaction options
5	Confirm	Run permission check, generate file, save a copy to the matter
6	Deliver	Download, email, send to portal, or send for e-signature
7	Record	Write an audit entry with user, time, format and settings

5. Technical Design

Component	Approach
Template engine	Merge fields, conditional sections and repeating blocks driven by matter and client data
Render service	Server-side rendering to a paged format for accurate preview and PDF output, so preview and export match
Preview delivery	Page images or a PDF viewer streamed to the browser; incremental updates for changed pages only
Conversion service	DOCX, PDF, PDF/A, XLSX and CSV generation in isolated workers
Job queue	Background processing for batch, large and scheduled exports with retry and status tracking
Caching	Cache rendered pages keyed by template version and data hash to speed repeat previews
Storage	Exports saved to encrypted object storage with version link to the source
API	Endpoints for create preview, update preview, request export, check job status and download

The key rule is that preview and export use the same rendering pipeline, so what the user approves is what the recipient receives.

6. Security and Privilege Controls

- Every preview and export checks the user's permissions on the matter and on each document.

- Ethical walls block preview and export of restricted matters for excluded users.
- Privileged content can be flagged, and exports containing it require confirmation.
- Download links are encrypted, time-limited and revocable.
- Full audit log of who previewed, exported or shared what, and when.
- Preview images are not stored in the browser cache after the session ends.

7. Non-Functional Requirements

Area	Requirement
Fidelity	Preview matches exported PDF in layout, fonts and pagination
Performance	Standard previews under 1 second; a 100-page export under 30 seconds; batches run in the background
Reliability	Failed jobs retry automatically and notify the user with a clear reason
Accessibility	Tagged PDFs, keyboard navigation and screen reader support in the preview panel
Scalability	Render workers scale with demand during month-end billing and filing peaks

8. Roadmap

Phase	Scope
Phase 1	Live template preview, validation panel, PDF and DOCX export, audit logging
Phase 2	Invoice and report previews, XLSX and CSV export, watermarks, filename patterns, presets
Phase 3	Version comparison, client view preview, redaction, metadata scrubbing, ZIP bundles
Phase 4	Bates numbering, PDF/A, batch and scheduled exports, e-signature and portal delivery

9. Acceptance Criteria

- Preview and exported PDF are visually identical for the same inputs.
- Missing required fields are flagged before export and can be fixed from the warning list.
- Redacted text cannot be recovered from the exported file.
- Users cannot preview or export documents outside their permissions.
- Every export is logged and a copy is saved to the matter.
- Batch exports complete in the background and notify the user when ready.

10. Risks and Mitigations

Risk	Mitigation
Preview differs from final output	Single shared rendering pipeline; automated visual comparison tests

Risk	Mitigation
Slow rendering on large documents	Page-level caching, incremental updates and background jobs
Leaking privileged or hidden data	Metadata scrubbing, permanent redaction, export confirmation and audit logs
Court format requirements vary	Configurable presets per court; verify against current local rules
Template errors break output	Template validation and test data scenarios before publishing

Note: This is a draft specification. Court and jurisdiction filing requirements change and should be confirmed before implementation.